

Abderrahman ES-SEBYITY

AI/ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building production RAG pipelines, fine-tuning LLMs, and deploying deep learning models. Strong full-stack background (React, Django, FastAPI). Master's thesis outperformed six published RAG baselines on multilingual benchmarks.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Keras, Scikit-learn, LLM Fine-Tuning (QLoRA), RAG, LangChain, NLP (SpaCy, Transformers), Computer Vision, Deep Learning, CNN

Languages: Python, JavaScript, TypeScript, Java **Data:** Pandas, NumPy, ChromaDB, MongoDB, PostgreSQL

Frameworks: FastAPI, Django, Flask, React, Next.js, Node.js, Express.js **Tools:** Git, Docker, Hugging Face, Unsloth

EXPERIENCE

AI/ML Engineer — Master's Thesis Feb 2026 – Jun 2026
Université Cadi Ayyad

- Designed an agentic RAG system answering Moroccan public-service questions in four languages (MSA, Darija, Arabizi, French), running locally on a single GPU with a 4B open-weights generator.
- Architected an 11-stage pipeline: hybrid retrieval (BM25 + BGE-M3 with RRF), cross-encoder reranking, LLM router, and plan-execute-reflect-synthesise agent loop with five deterministic tools.
- Built NLI verification scoring claims against evidence with per-claim citations, reducing hallucination.
- Outperformed six published baselines: 0.66 multi-hop success, 0.289 Arabizi F1 (next-best 0.034).

Frontend Developer Apr 2024 – Sep 2024
Integral Progress Technology

- Built dynamic pages for an ERP Quality Management module using React, TypeScript, TailwindCSS, and Mantine.
- Integrated REST APIs and managed global state with Redux Toolkit; built reusable component library.

Freelance Developer Jan 2023 – Sep 2023
Freelance

- Integrated the OpenAI API to auto-generate articles from keywords; deployed apps on cloud servers.
- Developed web scraping and automation scripts to streamline data extraction for clients.

Full-Stack Developer Nov 2022 – Dec 2022
FSJES Marrakech

- Built a diploma-tracking web app with Django REST Framework and React, serving real-time status updates.

PROJECTS

Agentic Multilingual RAG System — *Python, PyTorch, LangChain, ChromaDB, QLoRA, FastAPI*

- 11-stage RAG with hybrid retrieval, NLI claim verification; fine-tuned Mistral-Nemo 12B via QLoRA/Unsloth.

Automated Fact-Checking Pipeline — *Python, PyTorch, QLoRA, LangChain, ChromaDB, Unsloth*

- Fine-tuned Mistral-Nemo 12B SLM matching 91% of Gemini 2.5 Pro quality at 7x lower cost; RAG with verified sources.

Distributed Multi-Agent Anomaly Detection — *Python, XMPP, River, Streamlit*

- 20+ autonomous agents with online ML (HalfSpaceTrees), trust-based consensus in <200ms, 30% node-loss tolerance.

Brain Tumor Classification — *Python, TensorFlow, Keras, Flask, Docker*

- CNN using VGG16 transfer learning to classify MRI scans into four tumor categories.

EDUCATION

Master's in Data Science and Analytics Oct 2024 – Jul 2026
Université Cadi Ayyad — Polydisciplinary Faculty of Safi

Bachelor's in Computer Software Engineering Oct 2023 – Jun 2024
Higher School of Technology of Essaouira — Cadi Ayyad University